Linear systems and elimination

Sharp growth for unequal lower and upper bandwidths

Difficulty: challenging **Importance:** interesting to specialist

Status: open in the historical source; no resolution found in the bounded later search below.

Problem statement

Work in exact arithmetic over $\mathbb{C}$. For integers $p, q \geq 0$, define

$$\mathcal{B}_n(p,q) = \{A \in \mathbb{C}^{n\times n} : \det A \neq 0, \quad a_{ij} = 0 \text{ if } i-j > p \text{ or } j-i > q\}.$$

Gaussian elimination with partial pivoting (GEPP) chooses a largest-modulus entry in the active first column, moves its row to the first position, and forms the trailing Schur complement. For a permitted path π, let $S_1 = A, \ldots, S_n$ be its active matrices and define

$$\rho(A,\pi) = \frac{\max_{1\leq k\leq n} \|S_k\|_{\max}}{\|A\|_{\max}}, \qquad \|M\|_{\max} = \max_{i,j} |m_{ij}|.$$

Determine, for all $p \neq q$, the sharp bound independent of dimension,

$$G(p,q) = \sup_{n\geq 1+\max(p,q)} \ \sup_{A\in\mathcal{B}_n(p,q)} \ \sup_{\pi \text{ permitted by GEPP}} \rho(A,\pi).$$

Identify the least universal upper bound and matching examples approaching it. All tie choices are included. The bandwidth restrictions apply in the given ordering; no reordering before GEPP is allowed. Bandwidths are *at most* p and q; unequal pairs with one zero bandwidth are included. For $p \geq 1$, the equal-bandwidth case is known:

$$G(p,p) = 2^{2p-1} - (p-1)2^{p-2}.$$

This asks how unequal bandwidths affect worst-case element growth in banded linear solves. All unequal pairs constitute one problem.

References

N. J. Higham, *Accuracy and Stability of Numerical Algorithms*, second edition, SIAM (2002), Problem 9.15(a), p. 193; definitions and Theorem 9.11, pp. 172–173. Z. Bohte, *Bounds for Rounding Errors in the Gaussian Elimination for Band Systems*, J. Inst. Maths. Applics. 16 (1975), 133–142.

Status check

The book's problem was checked directly. Its wording omits the field; this entry adopts $\mathbb{C}$ from its referenced Theorem 9.11. Searches for *Higham 9.15 growth*, *growth factor lower bandwidth upper bandwidth*, and *Bohte Gaussian 1975* found no general unequal-bandwidth solution. Shah–Urschel's August 31, 2026 revision, Theorems 2.2–2.3, studies sparsity constraints without determining this extremal function. No recent explicit reaffirmation of this particular question's openness was located; the status rests on the original question and this bounded later-literature check.